

Customer Behavior Analysis Report

1. Project Overview

The objective of this project was to analyze customer transactions and customer behavior to identify customer segments, purchasing patterns, and churn risks. The analysis was carried out using Python libraries such as Pandas, Matplotlib, and Scikit-learn.

2. Dataset Information

Dataset: Ecommerce Customer Data

Total Records: 250,000

Total Features: 13

Key Attributes:

- Customer ID
- Purchase Date
- Product Category
- Product Price
- Quantity
- Total Purchase Amount
- Payment Method
- Age
- Gender
- Returns
- Churn

3. Data Cleaning

The following preprocessing steps were performed during data cleaning:

- Checked for missing values
- Filled missing values in the Returns column
- Removed duplicate records
- Converted Purchase Date into datetime format

4. Feature Engineering

New features were created to improve analysis and identify trends:

- Purchase Month
- Purchase Year

These features helped analyze seasonal purchasing behavior and yearly sales trends.

5. Purchase Pattern Analysis

The following analyses were performed:

- Product Category Sales Analysis
- Payment Method Usage Analysis
- Monthly Sales Trend Analysis

Key Findings:

- Some product categories generated significantly higher revenue.
- Customers preferred specific payment methods more frequently.
- Monthly sales trends varied across different time periods.

6. Customer Segmentation (RFM Analysis)

RFM analysis was performed using the following metrics:

- **Recency:** Days since the customer's last purchase
- **Frequency:** Number of purchases made by the customer
- **Monetary:** Total amount spent by the customer

These metrics helped identify customer value and purchasing behavior.

7. Customer Clustering

K-Means Clustering algorithm was applied with 4 customer clusters to group customers based on purchasing behavior and spending patterns.

8. Churn Analysis

Customer churn behavior was analyzed using the Churn column.

Findings:

- A subset of customers showed higher churn risk.
- At-risk customers generally had lower purchase frequency and lower spending behavior.

9. Recommendations

- Reward high-value customers through loyalty and reward programs.
- Re-engage at-risk customers using personalized promotions and discounts.
- Focus marketing campaigns on top-performing product categories.
- Reduce product returns through better product quality and accurate descriptions.
- Use customer segmentation for personalized marketing strategies.

10. Conclusion

The analysis successfully identified customer segments, purchasing trends, and churn risks.

These insights can help improve customer retention, increase business revenue, and support data-driven decision-making.

Cluster Profiles

Cluster	Segment
2	Best Customers
0	Loyal Customers
3	Regular Customers
1	At-Risk Customers

Cluster Statistics

Cluster	Recency	Frequency	Monetary
0	222.65	5.37	15606.01
1	467.29	2.20	4564.50
2	164.09	7.43	23288.13
3	293.93	3.80	9958.17

Final Submission Files

Submit the following project files:

- Customer_Analysis.ipynb
- cleaned_data.csv
- rfm_segments.csv
- ecommerce_customer_data_large.xsl
- README.md